

Bayesian Learning

Lecture 12 - Course summary

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- Wait, what did we actually do?

- **Subjective probability**
- **Bayesian Learning:** Bayes theorem

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{p(\mathbf{x})} \propto p(\mathbf{x}|\theta)p(\theta)$$

- The proportional constant is actually the **marginal likelihood**

$$p(\mathbf{x}) = \int p(\mathbf{x}|\theta)p(\theta)d\theta$$

used for model comparison.

Simple models - conjugate priors

- iid **Bernoulli model** - Beta prior
- iid **Poisson model** - Gamma prior
- iid **Exponential model** - Gamma prior
- iid **Normal known variance** - Normal prior

Multi-parameter models

■ Marginalization

$$p(\theta_1|\mathbf{x}) = \int p(\theta_1, \theta_2|\mathbf{x}) d\theta_2$$

- iid **normal with unknown variance** $N(\theta, \sigma^2)$. Marginal posterior for θ is student- t

$$\theta|\mathbf{x} \sim t\left(\mu_n, \frac{\sigma_n^2}{\kappa_n}, \nu_n\right)$$

- **Multinomial model** - Dirichlet prior
- **Dirichlet distribution** on unit simplex $\sum_{k=1}^K \theta_k = 1$.

- **Expert** elicitation
- **Other data** sources
- **Non-informative priors** (prior sample size). Bernoulli model:

$$\theta | \mathbf{x} \sim \text{Beta}(\alpha + s, \beta + f)$$

- **Jeffreys' prior** $p(\theta) = |I(\theta)|^{1/2}$
- **Hierarchical priors**. L2-regularization

$$p(\boldsymbol{\beta}, \sigma^2, \lambda) = p(\boldsymbol{\beta} | \sigma^2 \lambda) p(\sigma^2) p(\lambda)$$

- **Regularization** and **smoothness priors**

Prediction

- **(Posterior) predictive distribution** for iid data:

$$p(\tilde{y}|\mathbf{y}) = \int_{\theta} p(\tilde{y}|\theta)p(\theta|\mathbf{y})d\theta$$

- ▶ $p(\tilde{y}|\theta)$ is the model density for a new observation $\tilde{y}$ given θ
- ▶ $p(\theta|\mathbf{y})$ is the posterior density for θ given the training data $\mathbf{y}$

- With no training data: the **prior predictive distribution**

$$p(\tilde{y}) = \int_{\theta} p(\tilde{y}|\theta)p(\theta)d\theta$$

where we integrate of the parameters using the **prior** $p(\theta)$.

- Use prior predictive density to set the prior hyperparameters when the expert expresses prior beliefs about observable data:
 - ▶ $\mathbb{E}(\tilde{y})$
 - ▶ $\Pr(\tilde{y} > c)$
- Interactive example with Gamma prior for Poisson model.

Decisions

- Need to make a **decision/action** $a \in \mathcal{A}$ when **state of nature** θ is partially unknown.
- The **utility function** describes the consequence of taking action a when state of nature is θ

$$U(a, \theta)$$

- The utility function is subjective (also for non-Bayesians) and is determined by the decision maker.
- **Bayesian theory** gives a single universal decision rule: choose the action that **maximizes (posterior) expected utility**:

$$EU(a) = \int U(a, \theta) p(\theta | \mathbf{y}) d\theta$$

where $\mathbf{y}$ is some training data.

Posterior approximation

- Three ways to approximate the posterior:
 - ▶ **Normal approximation**
 - ▶ **Posterior sampling** (Gibbs, MH, HMC)
 - ▶ **Numerical integration**
- Normal approximation is fast, but is approximate unless we have a lot of data (infinite in theory).
- Posterior sampling will converge to true posterior if simulated long enough, but is approximate in finite time.
- Numerical integration is only a good option when the number of parameters is small.

Model evaluation and comparison

- Three main ways to compare models
 - ▶ **Posterior model probabilities** (**marginal likelihood**)
 - ▶ **Log Predictive Density Score** (LPDS/elpd)
 - ▶ **Leave-one-out (LOO)** LPDS/elpd
- Evaluate the **predictive density performance** on test data

$$p(\mathbf{y}_{\text{test}}|\mathbf{y}_{\text{train}}) = \int p(\mathbf{y}_{\text{test}}|\boldsymbol{\theta}, \mathbf{y}_{\text{train}})p(\boldsymbol{\theta}|\mathbf{y}_{\text{train}})d\boldsymbol{\theta}$$

- ▶ test data $\mathbf{y}_{\text{test}}$
 - ▶ posterior $p(\boldsymbol{\theta}|\mathbf{y}_{\text{train}})$ based on some training data.
- Often evaluated on the log scale: $\log p(\mathbf{y}_{\text{test}}|\mathbf{y}_{\text{train}})$.
- **Parameter uncertainty is included** in the model comparison by integrating with respect to the posterior $p(\boldsymbol{\theta}|\mathbf{y}_{\text{train}})$.

Model evaluation and comparison

■ Marginal likelihood:

- ▶ no $\mathbf{y}_{\text{train}}$ and $\mathbf{y}_{\text{test}}$ is all the data.
- ▶ Sensitive to the prior on the parameters $p(\boldsymbol{\theta})$.

■ Log Predictive Density Score:

- ▶ $\mathbf{y}_{\text{train}}$ and $\mathbf{y}_{\text{test}}$ is some partition of the data.
- ▶ K -fold cross-validation.

■ Leave-one-out (LOO)

- ▶ $\mathbf{y}_{\text{test}} = y_i$ and $\mathbf{y}_{\text{train}} = \mathbf{y}_{-i}$ (all data except obs i)
- ▶ n -fold cross-validation.
- ▶ Time-consuming: posterior sampling from n posteriors

$$p(\boldsymbol{\theta}|\mathbf{y}_{-i}) \text{ for } i = 1, \dots, n$$

- ▶ The **loo** package uses a importance sampling trick for speed.